

HiCARE: Stratified Calibration, Temporal Updating and Evaluation Protocols

A

Base Learners and Inputs

33 base features + 18 historical non-ARI facility-profile features

Profile window: 12 months preceding the target month

LightGBM

Learning rate 0.05 | 127 leaves
Minimum leaf size 80 | L2 penalty 5

XGBoost

Learning rate 0.05 | Maximum depth 8
Minimum child weight 5 | L2 penalty 5

Feature and row subsampling fractions: 0.8 | Maximum 4,000 rounds
Early stopping on a stratified 10% training holdout; patience 100 rounds

Member probabilities → logit scores $z_i^{(m)}$
Probabilities clipped to $[10^{-7}, 1 - 10^{-7}]$

B

Three-Step Stratified Calibration

2a Stratified Logistic Stacking

Child and COVID-19 interactions with member scores; C = 1000

$$z_i^A = \alpha + \sum_m a_m z_i^{(m)} + c_i \left(b + \sum_m b_m z_i^{(m)} \right) + v_i \left(d + \sum_m d_m z_i^{(m)} \right)$$

c_i : child indicator; v_i : COVID-19 indicator

2b Shallow Residual-Tree Correction

$$z_i^B = z_i^A + h(u_i, z_i^A)$$

8 leaves | Minimum leaf size 1,000 | Learning rate 0.03 | L2 penalty 20

Subsampling 0.8; maximum 500 rounds; 15% of records for early stopping; patience 30 rounds

u_i includes specified population, facility, regional and monthly variables

2c Marginal-Category Logistic Projection

$$p_i^H = \sigma \left(\beta_0 + \beta_I z_i^B + \sum_k \gamma_k g_k(i) \right)$$

Age, sex, ethnicity, urban/rural residence, sector and infection type; year included in-distribution
Unpenalised fit using all second-layer training records

$$\sum_i g_k(i) (y_i - p_i^H) \approx 0$$

Marginal constraints depend on the fitting sample and convergence; they are not guaranteed in the test set.

C

Four-Dimensional Monthly State Updating

Development: 2015–2021

Temporal test: 2022–2023

$$\phi_i = [1, z_i^H, v_i, c_i]^T$$

$$z_i^H = \text{logit}(p_i^H)$$

$$\text{logit}(p_{i,t}) = \phi_i^T \theta_t$$

$$\theta_t = \theta_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, qI), \quad q = 0.1$$

Previous month's state

Predict current-month risk

Current-month outcomes available

Laplace update

Next month

Configuration selected by forward validation in 2018–2021
Four-dimensional state and $q = 0.1$ selected by log loss

Retrospective simulation by admission month; reporting delays were not strictly simulated

D

Three Evaluation Protocols

① In-Distribution Nested Cross-Validation

5 outer folds: 4 training + 1 test



Within the training partition, hold out one fold in turn
Fit base learners on 3 folds → generate nested predictions for 1 fold
Fit the second layer on nested predictions → evaluate on the outer test fold
20 outer–inner combinations per base learner

② Temporal Holdout

Development in 2015–2021 → testing in 2022–2023

Compare static models, rolling windows and matched state updates

③ Province-Grouped Holdout

6 province-held-out folds; geographic identifiers removed

Without profiles: LightGBM + XGBoost + TabM

With profiles: LightGBM + XGBoost

Only profile-ensemble configurations evaluated; the full second layer was not run

Historical statistics were constructed before splitting; historical information is not fully isolated.
The geographic comparison changes both profiles and learner composition.

E

Reference Models and Evaluation Metrics

Reference models

LR-7: seven-variable logistic regression

EW: equal-weight ensemble of LightGBM, XGBoost and TabM

EW probabilities are averaged, then logistically recalibrated

Evaluation dimensions

Discrimination: AUROC, AUPRC (average precision)

Probability error: log loss, Brier score

Calibration: ICI, intercept, slope

42 subgroups: weighted mean and maximum $|\log(\text{O/E})|$

Within-group error: worst-group ECE among groups meeting sample-size criteria

Projection uses all second-layer training records and shares some outcomes with residual-tree early stopping; interpret each evaluation protocol separately.